Analysis of Pretrained Models for Pest Classification

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1 Introduction

Pests are a consistent problem that farmers, gardeners, and other agricultural professionals face. Damage from pests can not only negatively impact crop quality but can also lead to total crop loss. To minimize crop pest infestation and maximize pesticide efficiency and effectiveness, professionals need a method of identifying what the pest is, when to spray pesticides, what pesticides to utilize, and how much. The goal is to minimize the cost and use of pesticides while maintaining a healthy and plentiful agricultural area via pest control.

In the past, farmers have relied on human identification, but that can be tedious and unreliable. With advancements in technology, machine learning and deep learning models have been developed to assist with this task. Although existing pest classification models are able to obtain high accuracy, many are trained on specific datasets and struggle to generalize to new environments. These models can also be sensitive to real-world conditions, like background noise, and struggle to perform well when presented with image variability. Additionally, there are gaps in actual model architectures. There are not a lot of existing models that implement attention or transformers.

This project aims to provide greater understanding about existing methods and models to successfully identify pests and therefore provide helpful insight on what necessary actions to take in order to combat and prevent damage from infestations. Specifically, we will be analyzing the performance of two existing model designs (convolutional neural networks (CNNs) vs Visual Transformers (ViTs)) on multiple new datasets by using evaluation metrics such as classification accuracy, precision, recall, and F1-score for each pest class. We are providing a comparative analysis that will help reveal how well different architectures perform on real-world, noisy pest image datasets. In other words, which architecture generalizes better?

2 Related Work

Machine Learning to Deep Learning for Insect Classification

- Survey on Crop Pest Detection Using Deep Learning and Machine Learning Approaches [1]

- Leveraging deep learning for plant disease and pest detection: a comprehensive review and future directions [2]
- Insect Pest Image Recognition: A Few-Shot Machine Learning Approach including Maturity Stages Classification [3]

Before the deep learning era, pest identification relied on traditional machine learning and computer vision techniques using handcrafted features. Deep learning models, particularly CNNs and advanced variants, now dominate this task due to improved performance and feature learning capabilities. Since a decent amount of work already exists around creating new and improving model architectures and performance, rather than proposing new architectures, we aim to compare existing models and focus on how well they generalize across datasets and real-world conditions.

CNNs for Pest Classification

- A Convolutional Neural Network Algorithm for Pest Detection Using GoogleNet [4]
- Image Classification of Pests with Residual Neural Network Based on Transfer Learning [5]
- Large-Scale Insect Pest Image Classification [6]

According to these sources, CNN architectures, particularly with transfer learning, have historically performed well. CNN architectures such as ResNet and EfficientNet achieve high accuracy on specific datasets that focus on a single-domain pest. These, however, do not systematically evaluate how well the models generalize to new datasets comprising of images with real-world backgrounds that introduce noise. In contrast to previous work, our work addresses the gap using cross-dataset evaluation.

Vision Transformers for Pest Classification

- Insect recognition based on complementary features from multiple views. [7]
- An ensemble learning integration of multiple CNN with improved vision transformer models for pest classification [8]
- An image is worth 16×16 words: Transformers for image recognition at scale [9]
- Crop pest image recognition based on the improved ViT method [10]

The above sources discuss and evaluate transformer-based models for pest classification. They demonstrate improvements to capture global context. In contrast, they focus on designing new architectures rather than analyzing whether existing ViT models can generalize across pest datasets without architectural modification. We will compare ViT performance against the CNN baseline model on new data without proposing new architectures.

Generalization and Evaluation of Pretrained Models

– ImageNet Large Scale Visual Recognition Challenge [11]
– Do ImageNet Classifiers Generalize to ImageNet? [12]
– IP102: A Large-Scale Benchmark Dataset for Insect Pest Recognition [13]

Large-scale datasets such as ImageNet have played a critical role in advancing deep learning models for image classification and enabled the development of powerful pretrained models that are widely used across domains. However, there is still some uncertainty around how well they generalize to more specific tasks and datasets. Our work is different from these works because we directly evaluate generalization in the context of agricultural pest classification by testing pretrained models across different pest datasets, of varying sizes and class distributions, rather than a single benchmark dataset. We also want to further investigate how training these models on smaller, domain-specific pest data improves classification performance.

3 Methodology

3.1 Models

The two existing model designs evaluated in this project are a CNN and a ViT model architecture. CNN-based models have historically served as a strong baseline for pest classification tasks and are among the most well-studied architectures in this domain [7]. More recently, ViT models have emerged as a more competitive alternative when pretrained on large-scale datasets. There is also evidence that these models were experimented with as a method to improve results while simultaneously requiring fewer computational resources [9].

For the CNN baseline, we are using a pre-trained model called ResNet-50, where weights were initialized from ImageNet [11]. This model is widely used in prior pest classification studies, allowing us to compare directly to prior work. The model is sourced from PyTorch and TensorFlow/Keras and is adapted for our task by replacing the final classification layer with a fully connected layer to match the number of output classes. First, no additional training will be performed, and the pretrained weights are used directly for evaluation. The model is then adapted with fine-tuning layers to train on a subset of the pest datasets.

For the transformer-based approach, we will use the ViT-B/16 model pretrained on ImageNet, which follows the standard baseline from Dosovitskiy et al. [9] and is available on HuggingFace and PyTorch. Similar to the CNN, the final classification head is changed to match the number of classes. Again, first no training will be performed then the model will be fine-tuned on the target datasets.

3.2 Datasets

To ensure a meaningful comparison, we are conducting experiments on two to three new datasets. The first dataset, Dangerous Farm Insects (Kaggle), contains

1,591 relatively clean images with 15 classes (about 100 images per class). The second dataset, IP102 [13] (Kaggle), contains 102 classes and includes a wider range of 75,222 (on average 737 samples per class) noisy, real-world images, making it a more challenging benchmark. The third dataset, Pest Dataset V2 (Kaggle), consists of 34,200 images with 19 classes already split into train (1,500 per class), test (150 per class), and validation (150 per class) datasets. These datasets allow us to compare model performance for controlled and realistic conditions, and fine-tuning.

3.3 Data Preprocessing

All images are resized such that the shorter size is 256 pixels and then center-cropped to 224×224 pixels to match the input requirements of both ResNet-50 and ViT-B/16. The images are then converted to tensors and normalized using ImageNet mean and standard deviation. No augmentation is applied since the models are evaluated without training. Identical preprocessing is applied across both datasets to ensure a reasonable comparison.

4 Experiments

4.1 Experiment 1: Generalization Across Datasets

Main Purpose: To evaluate how well pretrained image classification models generalize across new pest classification datasets that differ in scale, class distribution, and background noise, we will load and test ResNet-50 and ViT-B/16 on three datasets: Dangerous Farm Insects, IP102, and Pest Dataset V2, using the same pre-processing described above. Results will be compared across datasets and model types (CNN vs ViT) to assess versatility and determine which architecture performs better for this task.

Evaluation Metrics: We will use classification accuracy for overall performance comparisons and F1-score to account for class imbalance in datasets such as IP102. Precision and recall will be computed per class and averaged for the F1-score. We will also include confusion matrices to identify potential common misclassifications between visually similar pest species, providing additional contextual insight.

4.2 Experiment 2: Fine-Tuning on Target Datasets

Main Purpose: The goal of this experiment is to explore the effect of fine-tuning pretrained models on domain-specific pest datasets and compare performance with zero-shot evaluation. Both ResNet-50 and ViT-B/16 will be fine-tuned on a subset of each dataset and evaluated on the remaining data, with the final classification layer adapted to match the dataset classes.

Evaluation Metrics: We will use accuracy, precision, recall, and F1-score for consistency and comparison with Experiment 1. Confusion matrices are included to analyze common misclassifications among pest species. We will compare performance between pretrained and fine-tuned models to quantify the impact of domain-specific training. We will also report training accuracy to assess differences in learning behavior between CNN and ViT models for pest classification tasks.

References

1. Chithambarathanu, M., Jeyakumar, M.: Survey on crop pest detection using deep learning and machine learning approaches. Multimedia Tools and Applications **82** (04 2023) 42277–42310
2. Shoaib, M., Sadeghi-Niaraki, A., Ali, F., Hussain, I., Khalid, S.: Leveraging deep learning for plant disease and pest detection: a comprehensive review and future directions. Frontiers in Plant Science **Volume 16 - 2025** (2025)
3. Gomes, J.C., Borges, D.L.: Insect pest image recognition: A few-shot machine learning approach including maturity stages classification. Agronomy **12**(8) (2022)
4. Yulita, I., Rambe, M., Sholahuddin, A., Prabuwono, A.S.: A convolutional neural network algorithm for pest detection using googlenet. AgriEngineering **5** (12 2023) 2366–2380
5. Li, C., Zhen, T., Li, Z.: Image classification of pests with residual neural network based on transfer learning. Applied Sciences **12**(9) (2022)
6. Doan, T.N.: Large-scale insect pest image classification. Journal of Advances in Information Technology **14**(2) (2023) 328–341
7. An, J., Du, Y., Hong, P., Zhang, L., Weng, X.: Insect recognition based on complementary features from multiple views. Scientific Reports **13**(1) (feb 2023) 2966
8. Xia, W., Han, D., Li, D., Wu, Z., Han, B., Wang, J.: An ensemble learning integration of multiple cnn with improved vision transformer models for pest classification. Annals of Applied Biology **182** (10 2022)
9. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., Houlsby, N.: An image is worth 16x16 words: Transformers for image recognition at scale (2021)
10. Fu, X., Ma, Q., Yang, F., Zhang, C., Zhao, X., Chang, F., Han, L.: Crop pest image recognition based on the improved vit method. Information Processing in Agriculture **11**(2) (2024) 249–259
11. Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., Huang, Z., Karpathy, A., Khosla, A., Bernstein, M., Berg, A., Fei-Fei, L.: Imagenet large scale visual recognition challenge. International Journal of Computer Vision **115** (09 2014)
12. Recht, B., Roelofs, R., Schmidt, L., Shankar, V.: Do imagenet classifiers generalize to imagenet? CoRR **abs/1902.10811** (2019)
13. Wu, X., Zhan, C., Lai, Y., Cheng, M.M., Yang, J.: Ip102: A large-scale benchmark dataset for insect pest recognition. In: IEEE CVPR. (06 2019) 8787–8796